



Networked Control System for a Micro-Integrated Coherent Optical Receiver

Introduction:

Optical receivers are key components in modern telecommunications infrastructure, that transform modulated light signals into electrical signals that can interface with electrical equipment.

The FIM24725 coherent receiver combines an incoming signal laser with a strong local-oscillator (LO) in an optical hybrid enabling phase-sensitive detection. Balanced photodiodes convert the interference into electrical I/Q signals [1]. The resulting electrical signal contains a “beat” frequency equal to the difference between the signal and LO frequencies, allowing extremely small wavelength differences to be measured and enabling recovery of both amplitude and phase information.

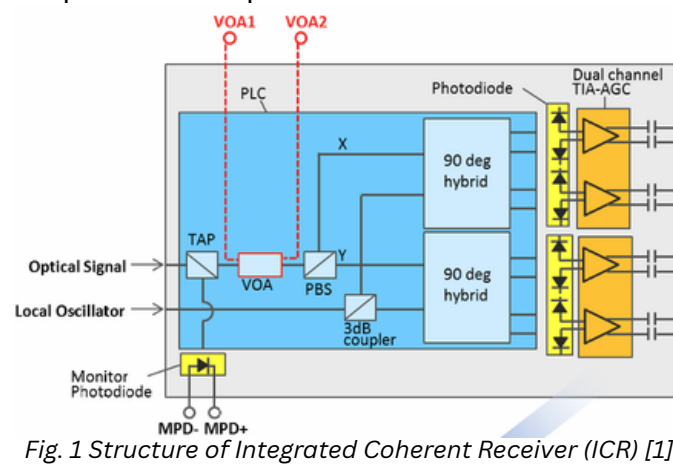


Fig. 1 Structure of Integrated Coherent Receiver (ICR) [1]

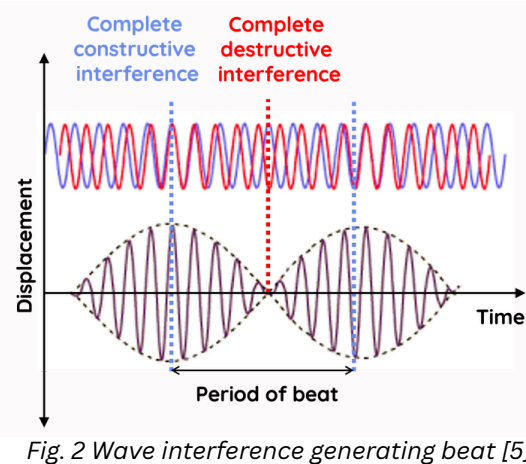


Fig. 2 Wave interference generating beat [5]

The receiver requires precise control of multiple supply rails and analogue control inputs to function, and strict adherence to sequencing constraints to avoid permanent hardware damage [1][2].

In the Trinity CONNECT laboratory, operation of this device was previously manual, error-prone, and difficult to reproduce. This project addresses these challenges by designing and implementing a lightweight, networked control system that enforces safe operating procedures while improving usability and repeatability.

Control System:

The system was designed to provide safe, state-aware, and remotely accessible control of a sensitive coherent receiver, ensuring correct sequencing, parameter validation, and reproducible operation.

- Safe hardware lifecycle management
- State machine enforcement
- Safe, validated, enforced control of all supply and control rails
- Real-time telemetry
- Concurrent access safety and fault handling
- Simple user-friendly interface
- Remote accessibility

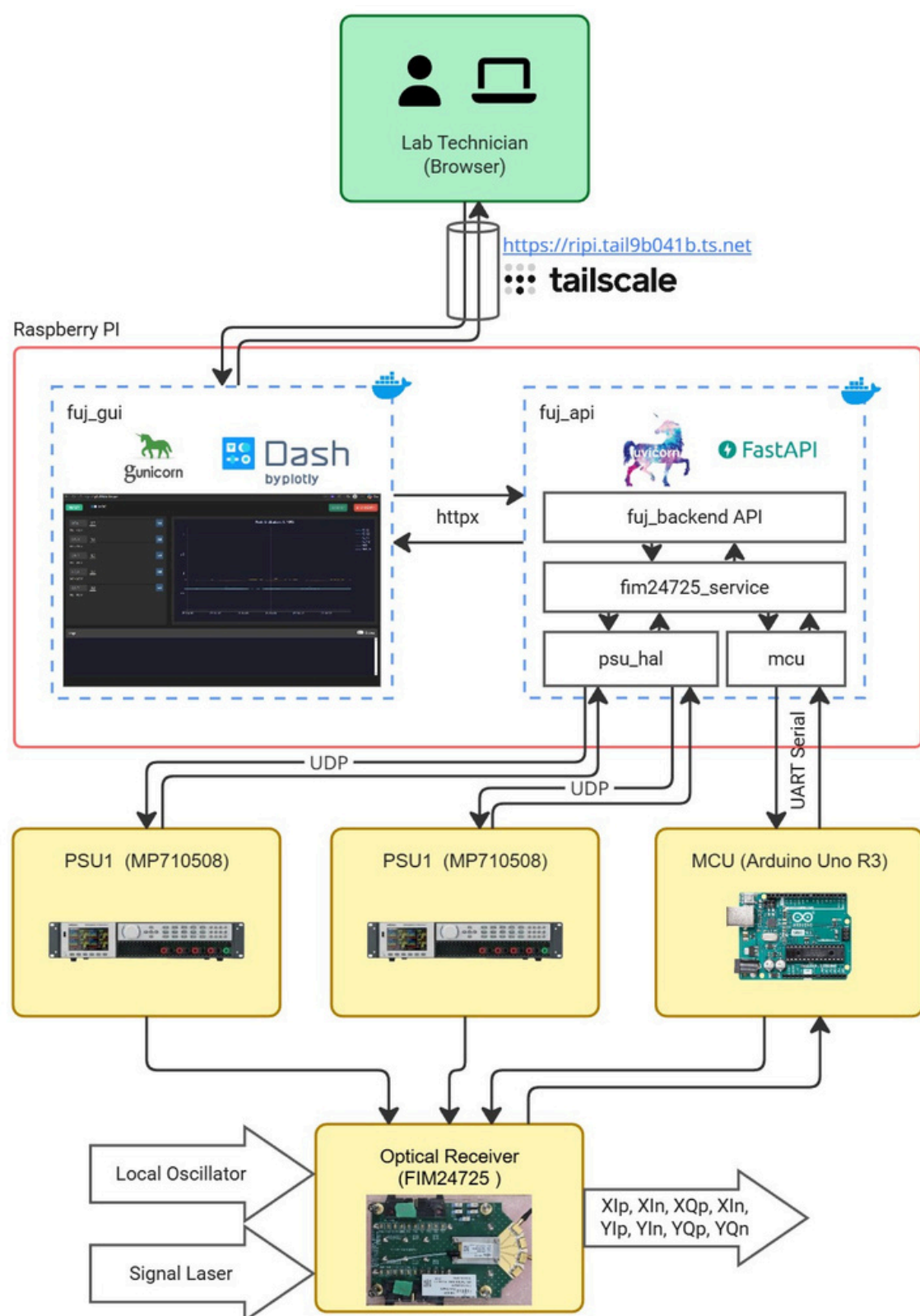


Fig. 3 Architecture diagram of ICR control system and components

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References:

- [1] Fujitsu Optical Components, “FIM24725 Micro-Integrated Coherent Receiver – Specification,” 2019.
- [2] Fujitsu Optical Components, “Application Note: Micro ICR Operation and Handling,” 2021.
- [3] Multicomp Pro, “MP710508 Programmable Power Supply User Manual,” 2020.
- [4] Agilent/Keysight, “Vector Signal Analyzer ESA Series – Operating Guide.”
- [5] Beats in Sound Waves Explained with Examples – HSC Physics [Internet]. Science Ready.

Experimental Setup:

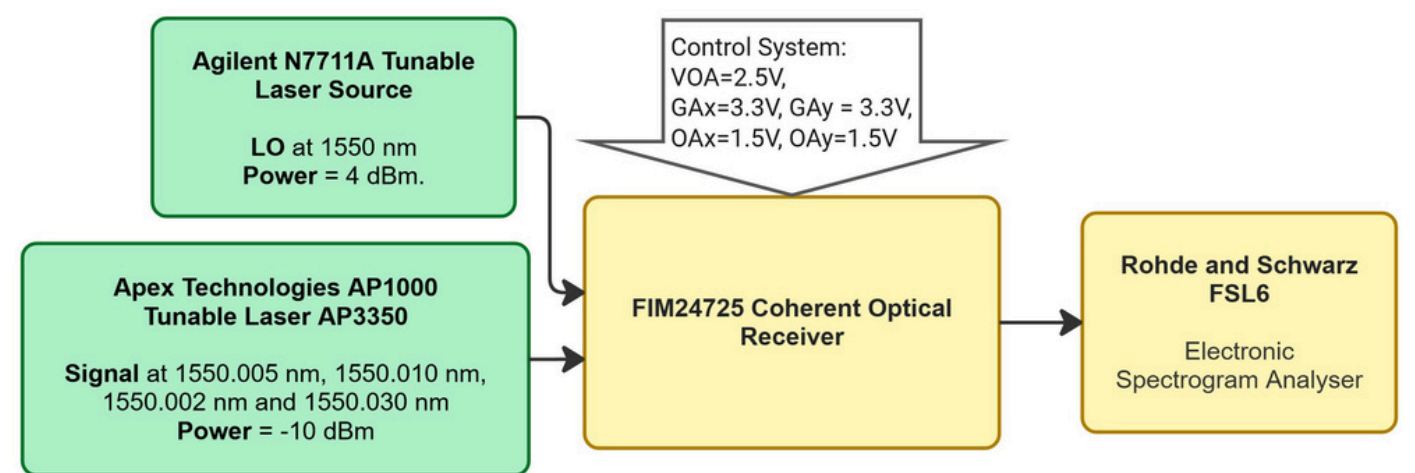


Fig. 4 Experimental setup signal flow

The coherent receiver mixes a weak optical signal and a strong local oscillator (LO) acting as a phase-stable reference. The difference in their optical frequencies produces a down-converted electrical intermediate frequency (IF), which the receiver’s photodiodes and transimpedance amplifiers recover as outputs XI(t), XQ(t), YI(t), YQ(t).

XI and YI outputs were measured with an electronic RF spectrogram analyser (ESA) to observe the beat tone between the lasers [4], which is approximated as: $\Delta f \approx \Delta \lambda \frac{c}{\lambda^2}$

Results:

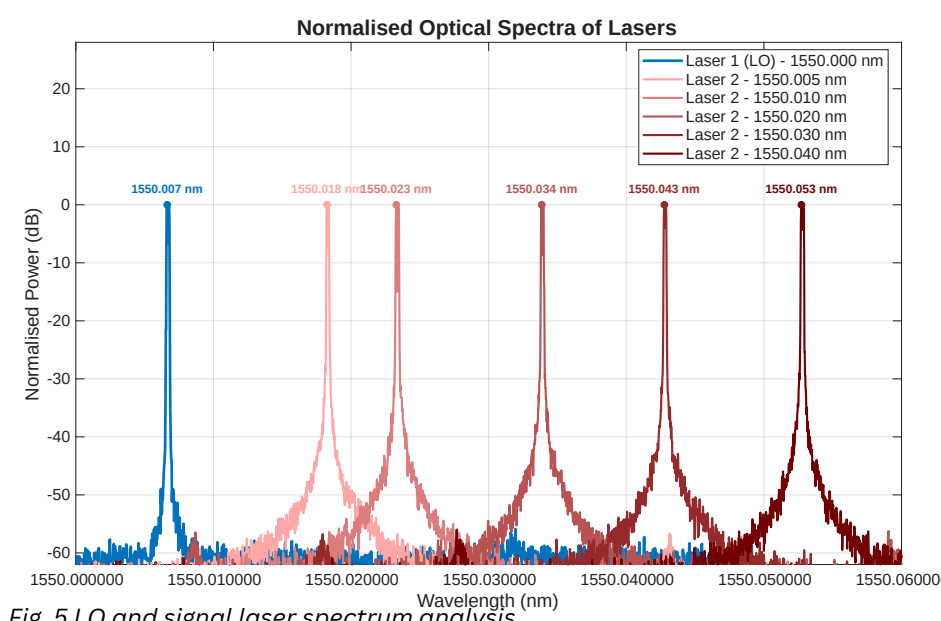


Fig. 5 LO and signal laser spectrum analysis

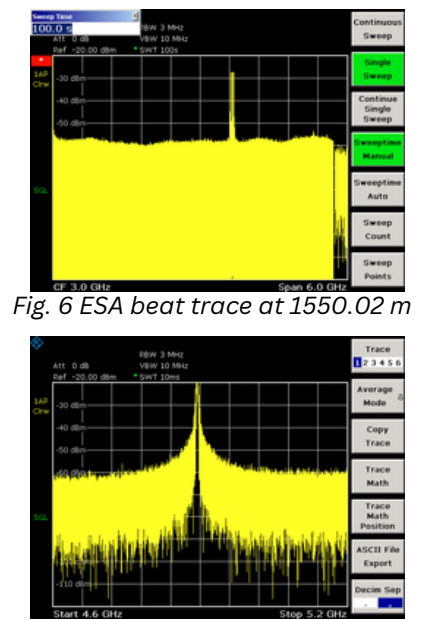


Fig. 6 ESA beat trace at 1550.02 nm

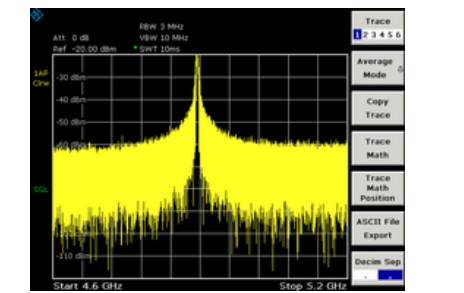


Fig. 7 Zoomed ESA beat at 1550.03nm

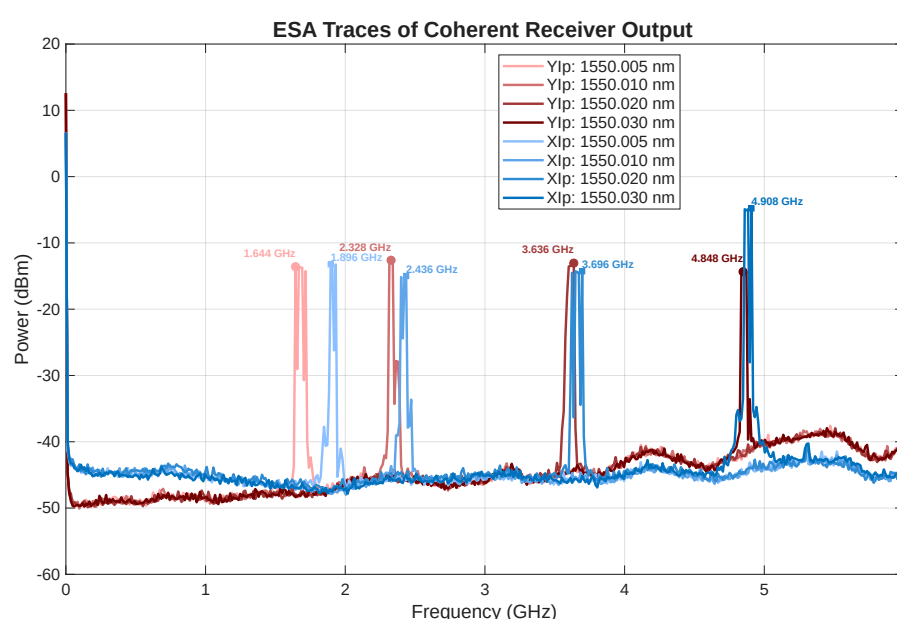


Fig. 8 Detected beat signals at different wavelengths

Table 1 Theoretical versus measured beat frequencies for YIp

Laser Setting	Measured λ (nm)	$\Delta\lambda$ (nm)	Theoretical Δf (GHz)	Measured Δf (GHz)	Error in Δf (GHz)
LO at 1550 nm	1,550.01	0	0	N/A	N/A
Signal at 1550.005 nm	1,550.02	0.011617	1.451	1.644	0.193
Signal at 1550.01 nm	1,550.02	0.016706	2.086	2.328	0.242
Signal at 1550.02 nm	1,550.03	0.027334	3.413	3.636	0.223
Signal at 1550.03 nm	1,550.04	0.036164	4.516	4.848	0.332
Signal at 1550.04 nm	1,550.05	0.046224	5.772	N/A	N/A

The presence of a well-defined and tunable beat frequency provides strong evidence of coherent detection. Unlike direct detection, which measures only optical intensity, the coherent receiver mixes the signal and LO fields, producing an electrical output at the difference frequency. The observed linear dependence of Δf on $\Delta\lambda$ therefore confirms correct operation of the coherent receiver.

Conclusion:

This project successfully met the design brief and implemented a functional control system for the coherent optical receiver, enabling stable operation of the experimental setup across multiple sessions. The operation of the coherent receiver was verified by measuring the beat frequency generated between a signal and local oscillator laser. The presence of a distinct and tunable electrical beat signal confirms that optical mixing occurred within the receiver, demonstrating successful coherent detection. While noticeable error was observed, this was attributed to practical limitations such as measurement uncertainty, laser drift, and instrumentation effects.

Future work will focus on integrating the system into a compact enclosure, developing a dedicated PCB, and improving overall usability and connectivity of the receiver. There is potential for exploring optimisation of receiver performance through control parameter tuning based on photodiode and signal feedback.

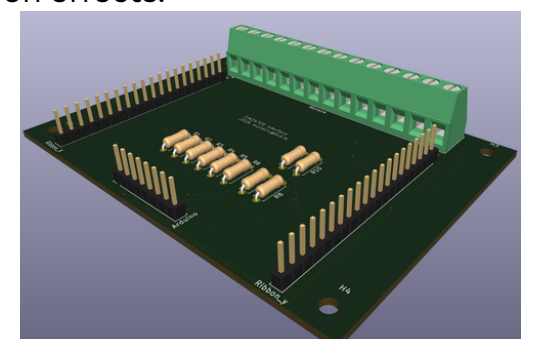


Fig. 9 PCB built for receiver, PSU, and MCU connection